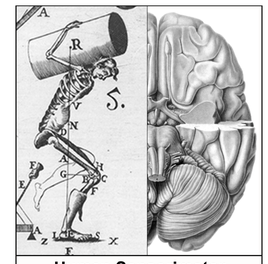
Deep brain stimulation of the subthalamic nucleus improves haptic perception in Parkinson's disease.

Aman JE[1,2], Abosch A[2,4,5], Bebler M[4], Lu C[1,2], Konczak J[1,2,3]

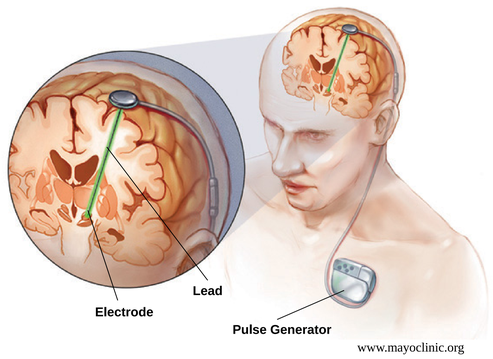
[1]School of Kinesiology, University of Minnesota, USA
[2]Center for Clinical Movement Science, University of Minnesota, USA
[3]Department of Neuroscience, University of Minnesota, USA
[4]Department of Neurosurgery, University of Minnesota, USA
[5]Institue for Engineering in Medicine, University of Minnesota, USA



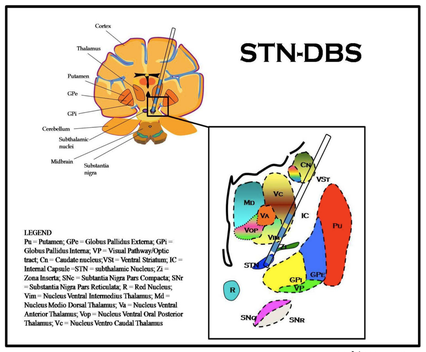
Human Sensorimotor Control Laboratory
hsc.umn.edu

Introduction:

Parkinson's disease (PD) is a neurodegenerative disease that affects the basal ganglia-thalamocortical pathway resulting in a progressive decline in motor function. An established treatment for the motor symptoms of PD is deep brain stimulation (DBS) of the subthalamic nucleus (STN). Mounting evidence suggests that PD is also associated with somatosensory deficits, specifically a loss of kinaesthetic and haptic precision, yet the effect of STN-DBS on sensory processing is largely unknown. Thus, this study investigated whether STN-DBS affects somatosensory processing by systematically examining the precision of haptic perception of object size.



www.mayoclinic.org



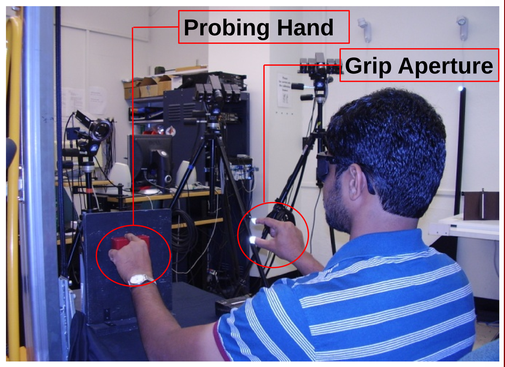
www.parkinson.org

Methods:

- 13 idiopathic PD patients with implanted STN-DBS
- Mean PD age: **58** ± 9.4 years
- Age-matched controls: **59** ± 7.1 years
- Disease duration: **10.5 years** (range: 5-20 years)
- Duration of DBS implant: **2.6 yrs** (range: 2mo-9yrs)
- DBS-OFF: UPDRS-III, 42±12; Hoehn & Yahr, 2.3
- DBS-ON: UPDRS-III, 27±9; Hoehn & Yahr, 1.7
- PD subjects tested in DBS-OFF & DBS-ON states with 12-hour medication wash-out.

Analysis

- Perceptual Judgments: Haptic perception discrimination thresholds (DT) derived from psychometric functions.
- Motor Judgments: Aperture error, calculated as the difference between the subject's grip aperture and actual height of the block being probed by the other hand.

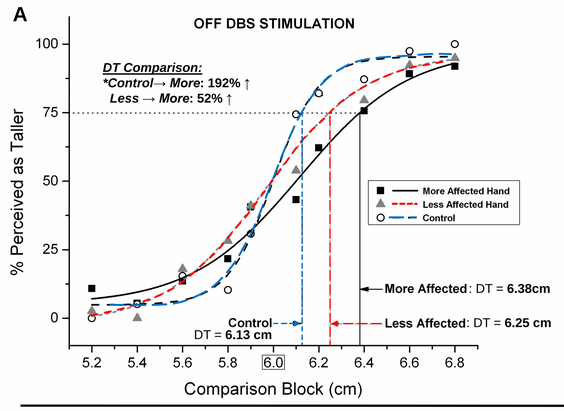


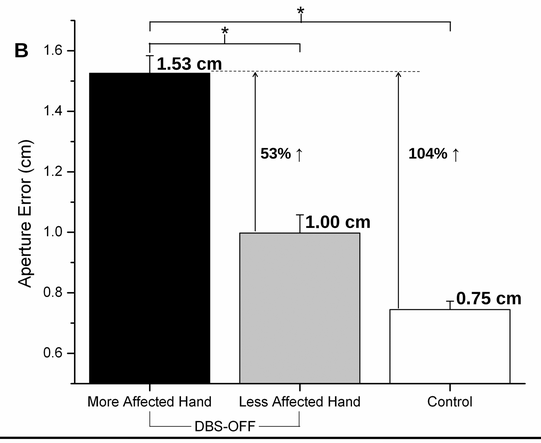
(6cm tall) **Reference Block** Presented & probed (4s) → (5.2-6.8cm) **Comparison Block** Presented & probed (4s) → Verbal forced choice ***perceptual* judgment** – "Which block was taller?" → While still grasping the comparison block – form Grip Aperture size for ***motor* judgment**

Results:

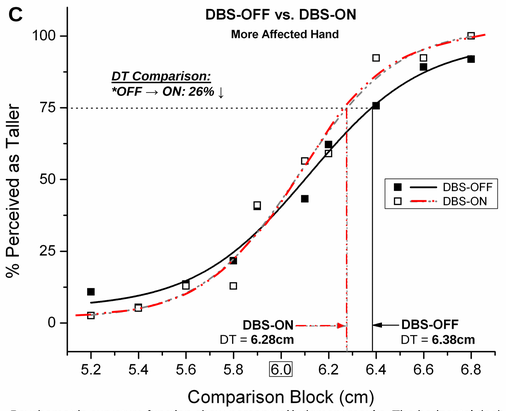
- **Parkinson's disease leads to a reduction in haptic precision.**
- **STN-DBS had a positive effect on proprioceptive function by improving haptic precision.**
- **PD patients probing with their *less* affected hand resulted in more precise perceptual *and* motor judgments than probing with the *more* affected hand.**

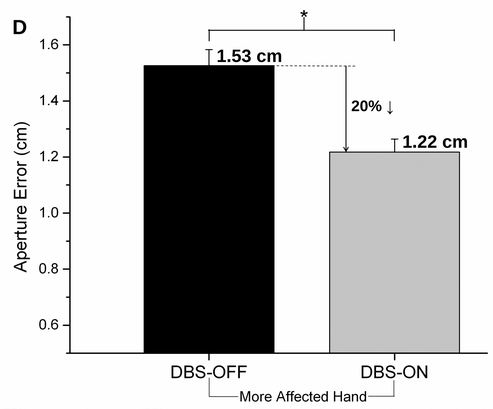
Haptic function deficits in PD patients





Effect of STN-DBS on haptic function in PD





Psychometric curvature function show *perceptual* judgment results. The horizontal dashed line indicates 75% perceived as taller (75% correct response rate) and the respective vertical lines indicate the DT. Bar graphs show results for the *motor* judgments with mean aperture error indicating the size deviation between the actual height of the block they are probing and their corresponding grip aperture intended to match the size of the block being probed. A) Results showing DT of PD patient's probing the block with their *more & less* affected hand in the DBS-OFF state and the control group. B) Mean aperture error. PD groups are labeled according to the hand used for probing, which would then mean their other hand was used to perform the (grip aperture) *motor* judgment. C) Results showing DT of PD patient's in the DBS-OFF vs. ON state, probing the block with their *more* affected hand. D) Mean aperture error for PD patients in the DBS-OFF vs. ON state, probing with their *more* affected hand. [Error bars indicate Standard Error].

Discussion:

- **PD is associated with deficits in haptic precision.**
- **STN-DBS (partially) alleviates haptic precision deficits.** STN-DBS may restore functional specificity to neurons within the basal ganglia-thalamocortical pathways , specifically within the STN by desynchronizing excessively synchronized neurons, or block preexisting erroneous signals from reaching cortical structures.
- **Motor deficits seen in PD may be a function of altered somatosensory processing, independent of goal-orientated motor function.** Improved *sensory* integration may then lead to improved motor function.